

Section 4K. Errors in LDA

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Classification Errors: LDA Theory (2)

What are the expected value of $|FP|$ and $|FN|$?

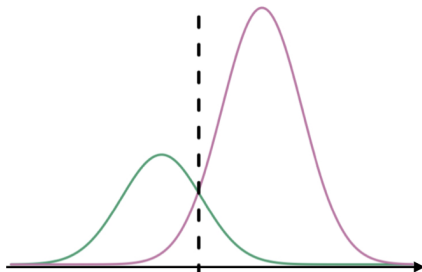
$$\begin{aligned}\mathbb{E}[|FP|] &= \sum_{i=1}^N \mathbb{E}_{X|Y} [\delta(C(X) - 1) \delta(Y)] \\ (\text{Total Expectation}) &= \sum_{i=1}^N \mathbb{E}_{X|Y} [\delta(C(X) - 1) 1|Y=0] \pi_0 + \mathbb{E}_X [0|Y=1] \pi_1 \\ &= \sum_{i=1}^N \int_{x \in \mathbb{R}} \delta(C(x) - 1) f_{X|Y}(x|0) \pi_0 dx \\ &= N \int_{x \geq \gamma} \pi_0 f_0(x) dx \\ \mathbb{E}[|FN|] &= (\text{similar derivation}) = N \int_{x < \gamma} \pi_1 f_1(x) dx\end{aligned}$$

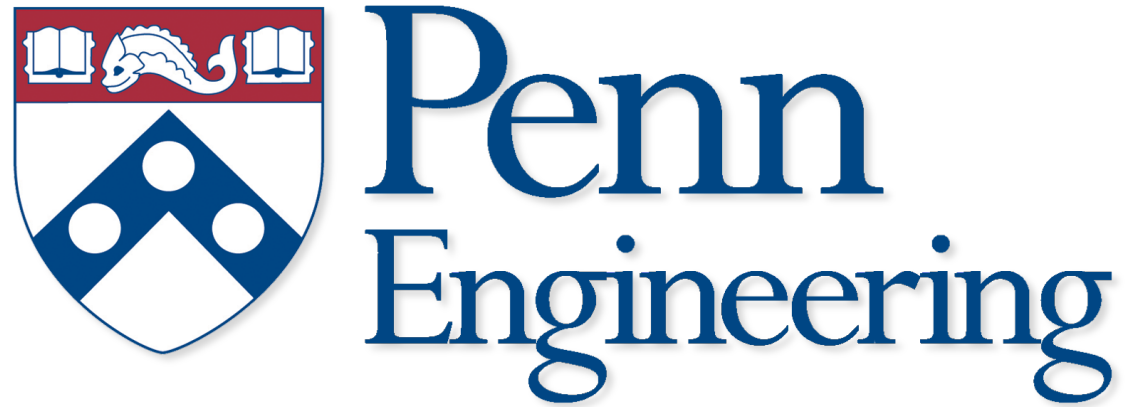
Classification Errors: LDA Theory (3)

Graphical Interpretation:

$$\mathbb{E}[|\text{FP}|] = N \int_{x \geq \gamma} \pi_0 f_0(x) dx$$

$$\mathbb{E}[|\text{FN}|] = N \int_{x < \gamma} \pi_1 f_1(x) dx$$





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